Applications

1

a Sine, the graph's initial value is at the midline.

b $I = \sin(90t)$

c 210 times

2

a 6 seconds

b $2500\ \text{mL}$

c $200\ \text{mL}$

d 60° per second

e $c = \frac{1}{2}$

3

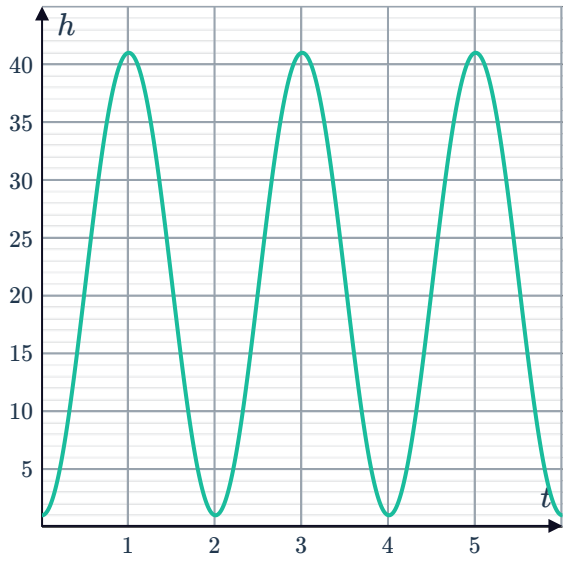
a

i No **ii** No **iii** No **iv** Yes

b

i 2 minutes **ii** $1\ \text{m}$ **iii** $41\ \text{m}$

c



d $h = -20\cos(180t) + 21$

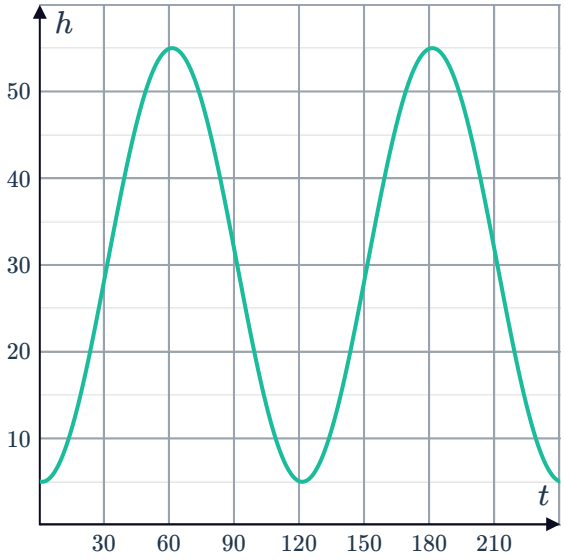
e $t = \frac{1}{2}$ minute

f The height of the passengers above or below the centre of the wheel at time t.

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4

a



b 55 m

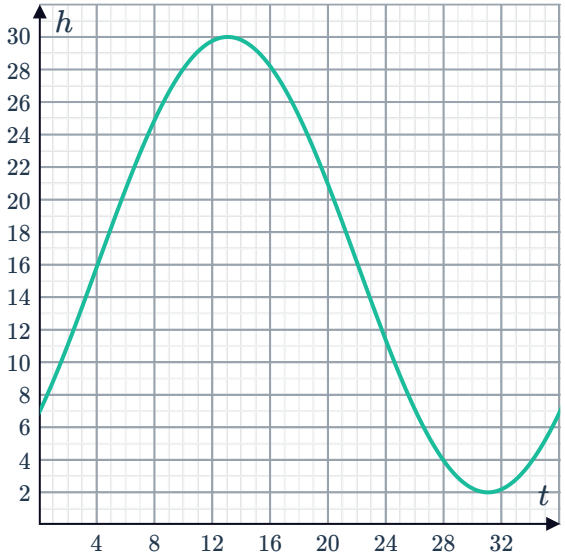
c 5 m

d 36.47 m

e 120 seconds

5

a



b 7.00 m

c $t = 34.82$ seconds

d 3

e 3.88 m

6

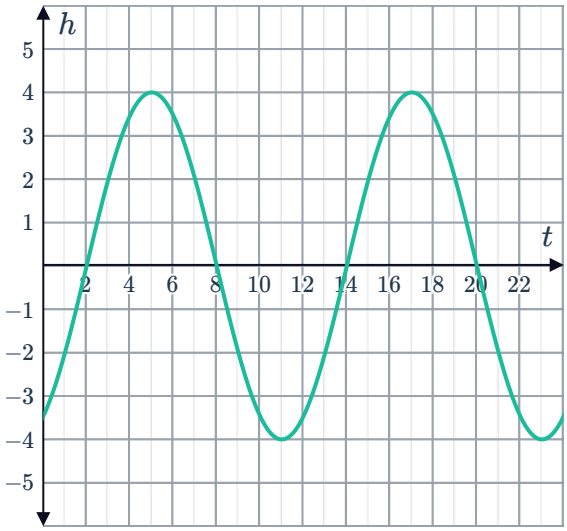
a -2 m

b 5:00 am

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c 6 hours

d



e 0 m

f 0.14 m

7

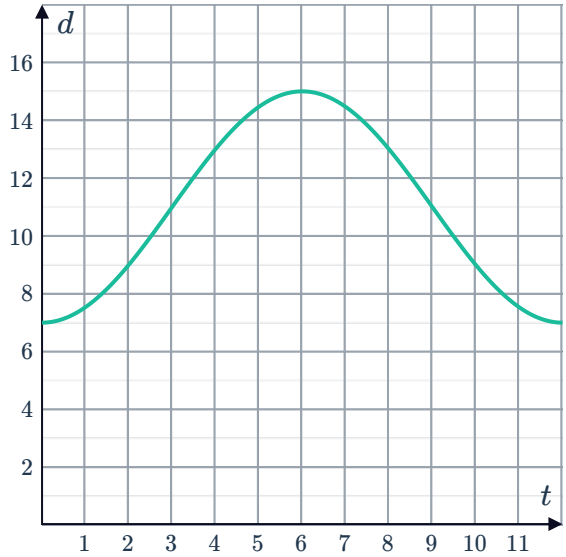
a

Time (t, hours)	0	3	6	9	12
Tide level (d, metres)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes **ii** No **iii** No **iv** No

f $d = -4\cos 30t + 11$

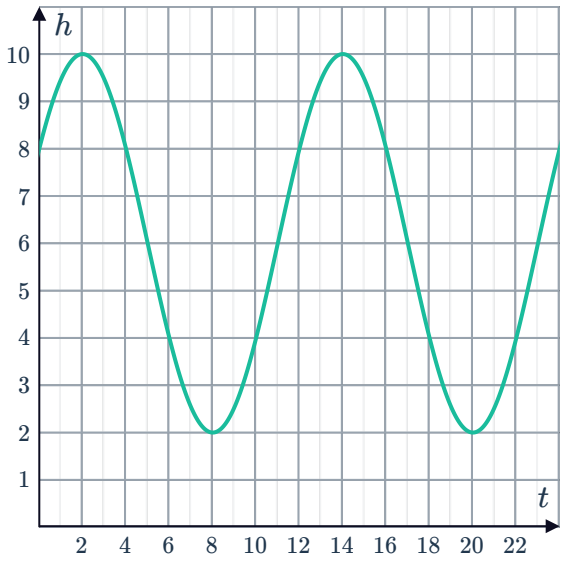
g 3.34 hours

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8

a $h(t) = 6 + 4\cos(30(t-2))$

b



c 2:00 am

d 8:00 am

9

a

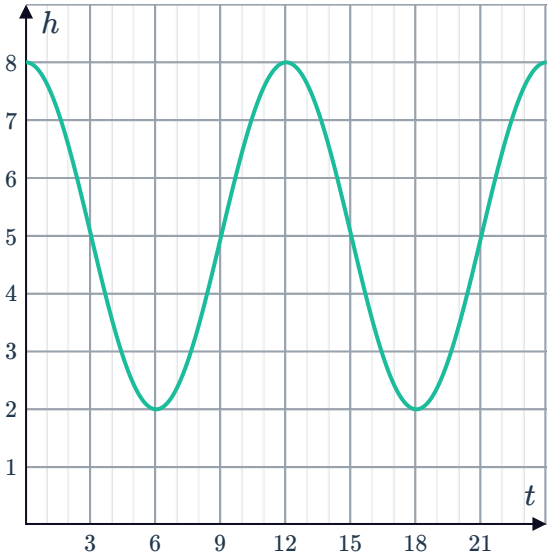
i $d = 5$ m

ii $a = 3$ m

iii $b = 30^\circ$ per hour

iv $c = 90^\circ$

b



c 6 and 18 hours after noon

10

a

i $a = 40$ cm

ii $d = 10$ cm

iii $c = \frac{1}{4}$

b $t = \frac{1}{3}$ second

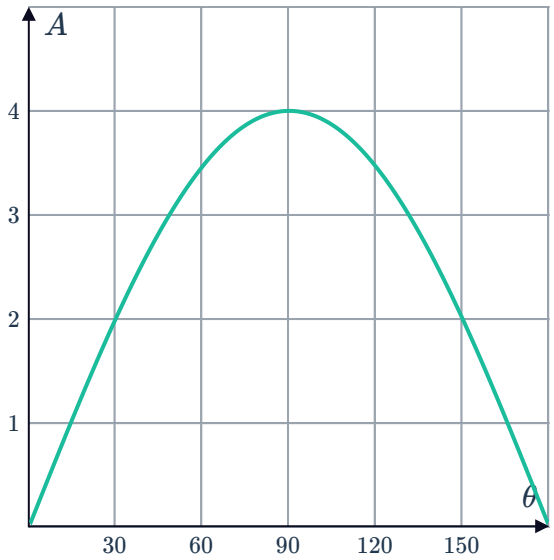
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11

a $A = 4\sin\theta$

b $0 < \theta < 180$

c



d 30°

12

a 5 cm to the right of the centre

b To the left

c 1 second

d 4 seconds

e $x = 5\cos(90t)$

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